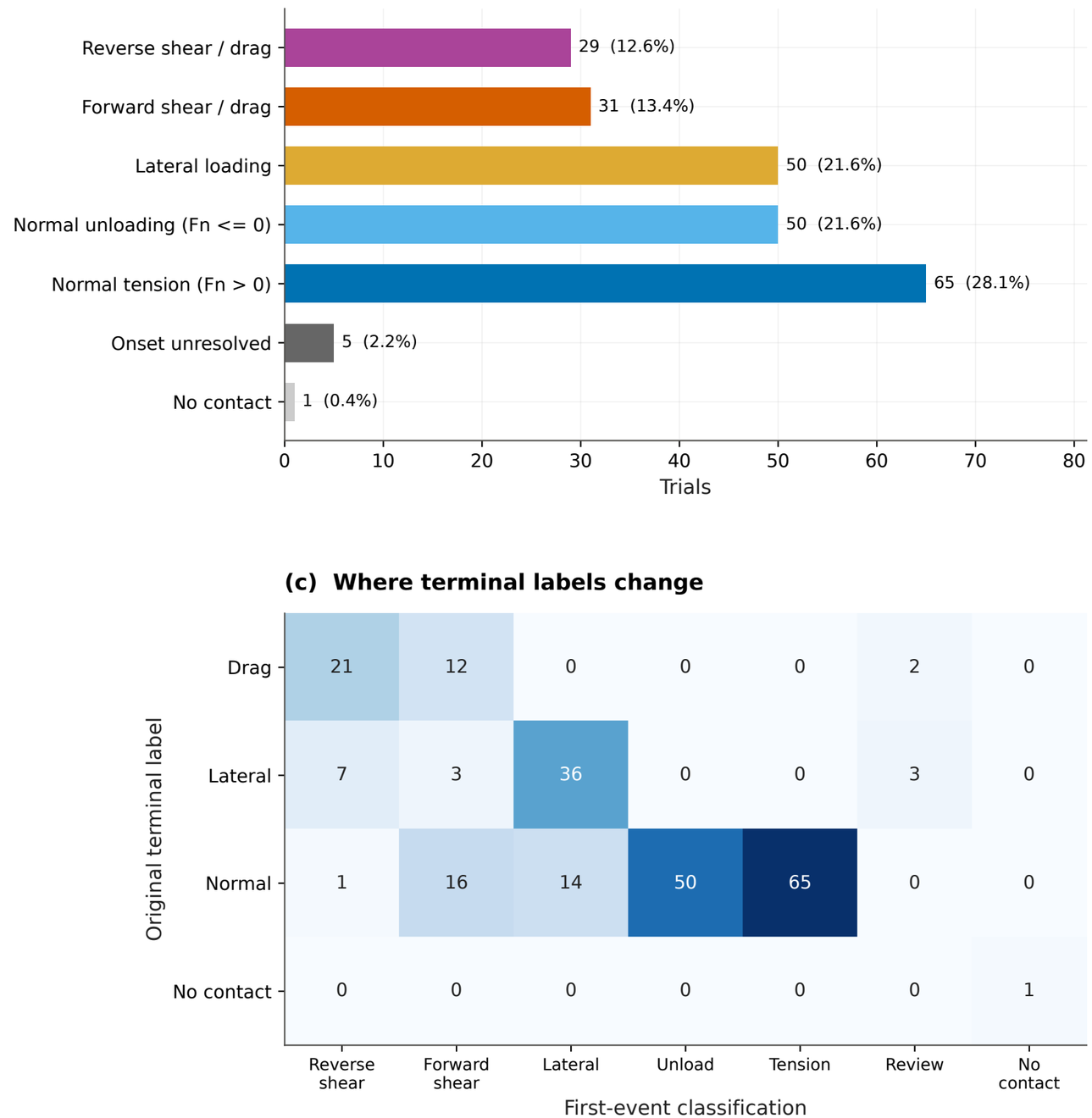
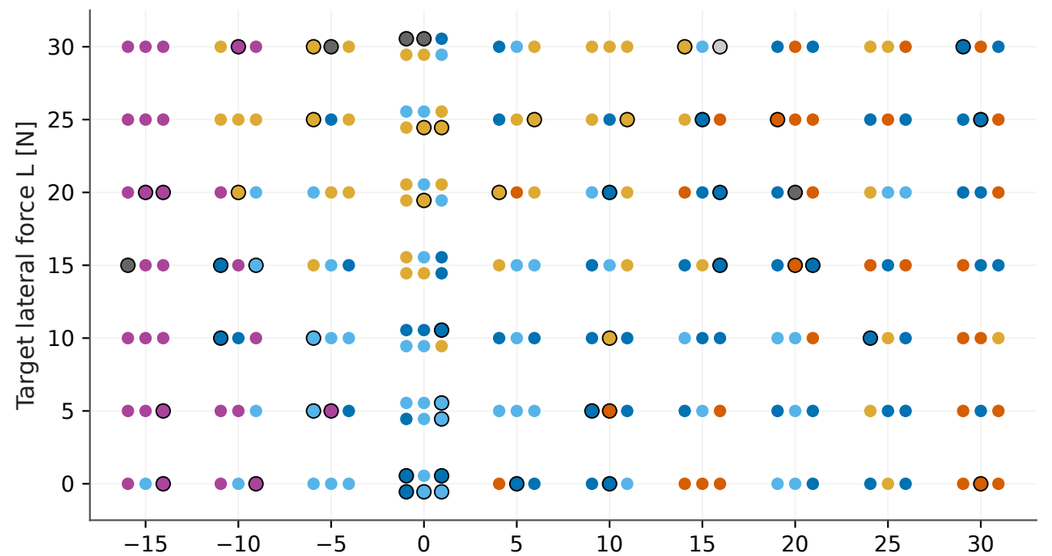
First-failure classification | 2026-09-22

231 trials | 3 N preload | first event across all loading stages | 93 re-engaged after first slip
Categories describe the loading stage; 52 traces need onset review.

(a) Inferred stage of the first event



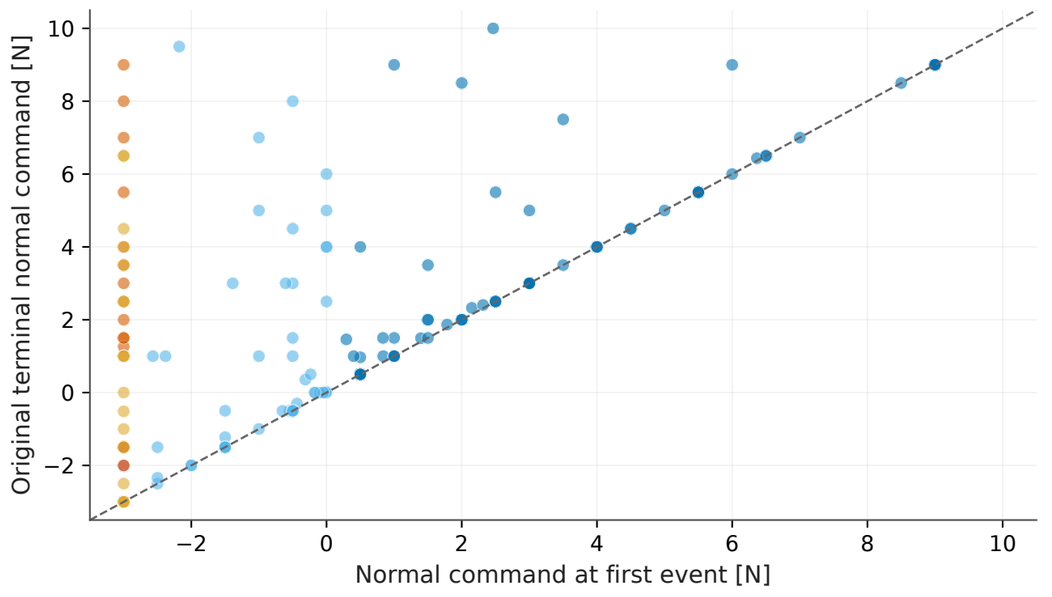
(b) All repetitions at each intended target



Left / middle / right = R1 / R2 / R3; black edge = review.
S=0 has both side labels (six trials per cell).

(c) Where terminal labels change

(d) Terminal loading can overstate first-slip capacity



Force and pose signals determine onset. Smooth creep and threshold-sensitive detections are provisional; these labels do not identify spine fracture or substrate damage.